

1. In the dependency Inversion principle, what should high level modules depend on?

Abstraction

2. What will be the output of the algorithm given below if $n = 5$?

ALGORITHM:

Step 1: Start

Step 2: Read the number n

Step 3: $j = 1$, $act = 1$

Step 4: Repeat Step 5 through Step 6 until $j > n$

Step 5: $act = j + 2 \times n - 2 \times act \times j$

Step 6: $j = j + 1$

Step 7: Print act

Step 8: Stop

- a) 25
- b) 2500
- c) 1500
- d) 750

3. How does few-shot prompting differ from zero-shot prompting?

- a. Few-shot requires fine-tuning, zero-shot doesn't
- b. Few-shot includes examples in the prompt, zero-shot does not
- c. Few-shot requires more memory
- d. Zero-shot is faster than few-shot

4. What is the total number of swaps if elements given below are sorted in descending order using bubble sort?

[15,12,45,67,89,32]

5. Which of the below given statement(s) is/are true?

- I. If Static Binding or Early Binding, the compiler can resolve the binding at compile time.
- II. All the static method calls are resolved at compile time because static methods are class methods, and they are accessed using the class name itself. Hence, there is no confusion for the compiler to resolve which version of the method is going to be called.

6. The checksum field in SCTP is bits and in TCP bits.

7. Which of the statements given below is/are false?

- I. To call more than two constructors of a class, we must use constructor chaining.
- II. We can place this () in all constructors in a class.
- III. In object creation, we are not allowed to call the same constructor more than once.

8. What will be the output of the flowchart given below for $k = 1$?

Start

Initialize $p = 2, j, k$

If $p < 5$?

- No $\rightarrow$ Stop
- Yes $\rightarrow$ next step

If $p > 5$ or $p - 1 < 5$?

- No $\rightarrow p = p + 1$
- Yes $\rightarrow$ next step

Set $j = p + 2$

Set $k = j - 2$

Print k

If $k < p$?

- Yes $\rightarrow$ Print k
- No $\rightarrow$ If $k < 10$ and $p < 8$?
 - Yes $\rightarrow$ Print Good
 - No $\rightarrow$ Print Nice

$p = p + 1$

Repeat

- a. 5 Nice 8 Good 13 Nice
- b. 2 Good 5 Nice 9 Nice
- c. 3 Good 6 Good 10 Nice
- d. 4 Good 7 Good 11 Nice

9. Your company is using a transformer-based generative AI for language translation. The translations sometimes fail to maintain the context of longer sentences. What could improve the model's performance?

- a. Switch to using an RNN-based translation model.
- b. Use a larger transformer model like GPT to handle longer context.
- c. Train the model on shorter sentences only.
- d. Manually translate long sentences and use the model for short phrases.

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Identify the type of binary tree.

- a. Degenerate Tree
- b. Full Binary Tree
- c. None of the options given
- d. Complete Binary Tree

11. What will be the value of variable *xxx* after going through the flowchart given below?

Input: *i* = 1, *j* = 3 and *k* = 4

- Start
- Read *i*, *j*, *k*, *x*
- Check if $i < j$?
 - Yes → check if $i < k$?
 - Yes → $x = i + k$
 - No → $x = j$
 - No → check if $j < k$?
 - Yes → $x = j - k$
 - No → $x = j$
- End

- a. 4
- b. 2
- c. 3
- d. 5

12. Given the Drink_info table, which of the query given below will give different output?

Drink_info(drink_name, cost, carbs, color, ice, calories)

drink_name	cost	carbs	color	ice	calories
pepsi	3	8.4	red	Y	33
lime	2.5	3.2	green	Y	24
blue moon	2.5	5.4	blue	Y	12
soda	3.8	4.7	red	N	19
indiansummer	2.8	7.2	brown	N	30
coca cola	3.2	32.1	red	N	135
spice	5.5	42.5	orange	Y	171
Miranda	3	8.4	yellow	Y	33
fruity	4.5	8.4	yellow	N	32

QUERIES:

I. SELECT drink_name, color FROM drink_info WHERE carbs < 4 AND ice='Y';

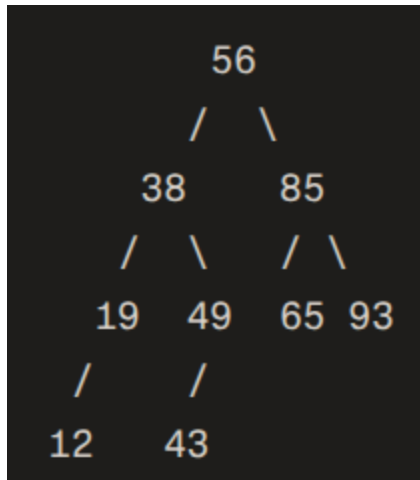
II. SELECT drink_name, color FROM drink_info WHERE carbs < 4 OR ice='Y';

III. SELECT drink_name, color FROM drink_info WHERE carbs < 4;

IV. SELECT drink_name, color FROM drink_info WHERE carbs < 4 AND ice='Y'
GROUP BY drink_name;

- a. Only (III)
- b. Only (I) and (IV)
- c. Only (I)
- d. Only (I) and (II)

13. Which one of the given options will be the new root of the below tree after deletion of 56?



- a. 38
- b. 85
- c. 43
- d. 49

14. You are tasked with writing a program that processes a string *S* of card representations and calculates a count value based on specific rules.

COUNTING RULES:

- Numeric cards (2-6) count as +1.
- Numeric cards (7-9) count as 0.
- Numeric card (10) counts as -1.
- Face cards (J, Q, K) count as -1.
- Ace (A) counts as -1.

DESCRIPTION:

Read the input from STDIN and print the output to STDOUT. Do not print arbitrary strings anywhere in the program, as these contribute to the output and test cases will fail.

CONSTRAINTS:

1. The length of string *S* > 0.

2. The string S can contain only 'A', 2-10, 'J', 'Q', and 'K' (only uppercase alphabets are allowed).
3. Card values in the string may repeat.

INPUT FORMAT:

The input will contain a string S that represents card values. The string S contains space-separated card representations.

OUTPUT FORMAT:

Output should have an integer representing the total count value calculated based on the rules described above.

SAMPLE INPUT 1:

5 9 10 3 J A 4 8 5

SAMPLE OUTPUT 1:

1

Explanation 1:

Given S = "5 9 10 3 J A 4 8 5".

Card	Count Value
5	+1
9	0
10	-1
3	+1
J	-1
A	-1
4	+1
8	0
5	+1

Total Count Value = 1

So the output is printed as **1**.

Sample Input 2:

J J Q K A A 2 2 4 9 10

Sample Output 2:

-4

Explanation 2:

Given S = "J J Q K A A 2 2 4 9 10".

Card	Count	Value
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J		-1
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J		-1
---	--	----

Q		-1
---	--	----

K		-1
---	--	----

A		-1
---	--	----

A		-1
---	--	----

2		+1
---	--	----

2		+1
---	--	----

4		+1
---	--	----

9		0
---	--	---

10		-1
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Total Count Value = -4

15, Draw a solution to a coding-decoding problem where one **"WORD"** is coded into another word. Analyze the following examples and write a program that accepts a word as input and prints its encoded text as output.

Examples Table

Sl.No	INPUT	OUTPUT
1	HELLO	1610242430
2	WORLD	463036248
3	OPEN	30321028
4	CLOSE	624303810
5	ZERO	52103630
6	A	2
7	Z	52
8	AZ	252

Instructions

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings anywhere in the program, as these contribute to the standard output, and test cases will fail.

Constraints

1. Length of the input "WORD"
2. Only English alphabets are allowed in the "WORD".
3. Only uppercase letters are allowed in the "WORD".

Input Format

The first line of input contains a "WORD" that is to be coded into a specific code.

Output Format

The first line of output should consist of the specific code which denotes the corresponding word coded version.

Sample Input 1

HELLO

Sample Output 1

1610242430

Sample Input 2

STREAM

Sample Output 2

38403610226